

Simon Oddershede Gregersen

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Employment

2026 – **CISPA Helmholtz Center for Information Security**. Tenure-track faculty.
2024 – 2026 **New York University**. Postdoctoral fellow. Advised by Joseph Tassarotti.
2023 – 2024 **Aarhus University**. Postdoctoral researcher. Advised by Lars Birkedal.
2021 **Huawei Technologies R&D (UK), Edinburgh**. Research intern. Advised by Dan Ghica.
2017 **Aarhus University**. Center for Advanced Software Analysis. Student programmer.
2015 – 2017 **CCI Europe A/S (now Stibo DX)**. Student software developer.

Education

2017 – 2023 **PhD in Computer Science**. Aarhus University.
Advised by Amin Timay and Lars Birkedal
Thesis: Higher-Order Separation Logic for Distributed Systems and Security
2017 – 2020 **MSc in Computer Science**. Aarhus University
2014 – 2017 **BSc in Computer Science**. Aarhus University

Publications

- [12] Li, K. H., Aguirre, A., **Gregersen, S. O.**, Haselwarter, P. G., Tassarotti, J., Birkedal, L., “Modular Reasoning about Error Bounds for Concurrent Probabilistic Programs”. In: *Proc. ACM Program. Lang.* 9.ICFP (Aug. 2025). DOI: [10.1145/3747514](https://doi.org/10.1145/3747514).
- [11] **Gregersen, S. O.**, Agarwal, C., Tassarotti, J., “Logical Relations for Formally Verified Authenticated Data Structures”. In: *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security, CCS 2025, Taipei, Taiwan, October 13-17, 2025*. 2025, pp. 1394–1408. DOI: [10.1145/3719027.3744801](https://doi.org/10.1145/3719027.3744801).
- [10] Haselwarter, P. G., Li, K. H., Aguirre, A., **Gregersen, S. O.**, Tassarotti, J., Birkedal, L., “Approximate Relational Reasoning for Higher-Order Probabilistic Programs”. In: *Proc. ACM Program. Lang.* 9.POPL (2025). DOI: [10.1145/3704877](https://doi.org/10.1145/3704877).
- [9] Haselwarter, P. G., Li, K. H., de Medeiros, M., **Gregersen, S. O.**, Aguirre, A., Tassarotti, J., Birkedal, L., “Tachis: Higher-Order Separation Logic with Credits for Expected Costs”. In: *Proc. ACM Program. Lang.* 8.OOPSLA2 (2024). DOI: [10.1145/3689753](https://doi.org/10.1145/3689753).
- [8] **Gregersen, S. O.**, Aguirre, A., Haselwarter, P. G., Tassarotti, J., Birkedal, L., “Almost-Sure Termination by Guarded Refinement”. In: *Proc. ACM Program. Lang.* 8.ICFP (2024). DOI: [10.1145/3674632](https://doi.org/10.1145/3674632).
- [7] Aguirre, A., Haselwarter, P. G., de Medeiros, M., Li, K. H., **Gregersen, S. O.**, Tassarotti, J., Birkedal, L., “Error Credits: Resourceful Reasoning about Error Bounds for Higher-Order Probabilistic Programs”. In: *Proc. ACM Program. Lang.* 8.ICFP (2024). DOI: [10.1145/3674635](https://doi.org/10.1145/3674635).

- [6] Timany, A., **Gregersen, S. O.**, Stefanescu, L., Hinrichsen, J. K., Gondelman, L., Nieto, A., Birkedal, L., “Trillium: Higher-Order Concurrent and Distributed Separation Logic for Intensional Refinement”. In: *Proc. ACM Program. Lang.* 8.POPL (2024). doi: [10.1145/3632851](https://doi.org/10.1145/3632851).
- [5] **Gregersen, S. O.**, Aguirre, A., Haselwarter, P. G., Tassarotti, J., Birkedal, L., “Asynchronous Probabilistic Couplings in Higher-Order Separation Logic”. In: *Proc. ACM Program. Lang.* 8.POPL (2024). doi: [10.1145/3632868](https://doi.org/10.1145/3632868).
- [4] **Gregersen, S. O.**, Bay, J., Timany, A., Birkedal, L., “Mechanized Logical Relations for Termination-Insensitive Noninterference”. In: *Proc. ACM Program. Lang.* 5.POPL (2021). doi: [10.1145/3434291](https://doi.org/10.1145/3434291).
- [3] Gondelman, L., **Gregersen, S. O.**, Nieto, A., Timany, A., Birkedal, L., “Distributed Causal Memory: Modular Specification and Verification in Higher-Order Distributed Separation Logic”. In: *Proc. ACM Program. Lang.* 5.POPL (2021). doi: [10.1145/3434323](https://doi.org/10.1145/3434323).
- [2] Krogh-Jespersen, M., Timany, A., Ohlenbusch, M. E., **Gregersen, S. O.**, Birkedal, L., “Aneris: A Mechanized Logic for Modular Reasoning about Distributed Systems”. In: *Programming Languages and Systems - 29th European Symposium on Programming, ESOP 2020, Held as Part of the European Joint Conferences on Theory and Practice of Software, ETAPS 2020, Dublin, Ireland, April 25-30, 2020, Proceedings.* 2020. doi: [10.1007/978-3-030-44914-8_13](https://doi.org/10.1007/978-3-030-44914-8_13).
- [1] **Gregersen, S. O.**, Thomsen, S. E., Askarov, A., “A Dependently Typed Library for Static Information-Flow Control in Idris”. In: *Principles of Security and Trust - 8th International Conference, Held as Part of the European Joint Conferences on Theory and Practice of Software, ETAPS 2019, Prague, Czech Republic, April 6-11, 2019, Proceedings.* 2019. doi: [10.1007/978-3-030-17138-4_3](https://doi.org/10.1007/978-3-030-17138-4_3).

Awards

2024	ICFP Distinguished Paper Award for the paper “Error Credits: Resourceful Reasoning about Error Bounds for Higher-Order Probabilistic Programs”.
2024 – 2026	Carlsberg Foundation Internationalization Fellowship (CF23-0791) DKK 1,020,000 ($\approx$ USD 150,000).

Academic Service

Organizing

- Artifact evaluation co-chair, [ACM SIGPLAN Conference on Object-Oriented Programming, Systems, Languages, and Applications \(OOPSLA\) 2026](#)

Program Committees

- [Workshop on Foundations of Computer Security \(FCS\) 2026](#)
- [ACM SIGPLAN Conference on Programming Language Design and Implementation \(PLDI\) 2026](#)
- [Workshop on Recent Advances in Concurrency and Logic \(RADICAL\) 2025](#)

External Reviewing

[JFP](#) (2025), [TCS](#) (2024), [TOSEM](#) (2023), [LICS](#) 2025, [LICS](#) 2024, [CCS](#) 2022, [POPL](#) 2022, [OOPSLA](#) 2022, [ICFP](#) 2022, [CSF](#) 2022, [CSF](#) 2021, [ESOP](#) 2020

Selected Talks

- *Building Extensible Program Logics through Effect Handlers.*
Invited talk at the MPI-SWS Foundations of Programming group seminar (6 February, 2026)

- *Logical Relations for Formally Verified Authenticated Data Structures.*
Contributed talk at the **New England Systems Verification Day** (2 October, 2025).
- *Logical Relations for Formally Verified Authenticated Data Structures.*
Contributed talk at the **Iris Workshop** (2 June, 2025).
- *Logical Relations for Formally Verified Authenticated Data Structures.*
Contributed talk at the **New Jersey Programming Languages and Systems Seminar** (9 May, 2025).
- *Logical Relations for Formally Verified Authenticated Data Structures.*
Invited talk at the VU Amsterdam PLSec reading group (16 April, 2025).
- *Trillium: Intensional Refinement in Higher-Order Separation Logic.*
Contributed talk at the **New England Systems Verification Day** (16 April, 2024).
- *Asynchronous Probabilistic Couplings in Higher-Order Separation Logic.*
Invited talk at the Bristol Programming Languages Research group seminar (19 July, 2023).
- *Asynchronous Probabilistic Couplings in Higher-Order Separation Logic.*
Contributed talk at **VeriProP** (17 July, 2023).
- *Trillium: History-Sensitive Refinement in Separation Logic.*
Contributed talk at **The Second Iris Workshop** (3 May, 2022).
- *Mechanized Logical Relations for Termination-Insensitive Noninterference.*
Invited talk at the Chalmers ProgLog/Security seminar (4 November, 2020).